

Python Learning Roadmap with Data Warehousing Concepts For Data Analyst, Data Engineer, BIE, and Cloud Roles

BEGINNER LEVEL

1. Python installation and environment setup
2. Python syntax and code structure
3. Variables and data types
4. Type casting and basic operators
5. Input and output operations
6. Strings and string operations
7. Lists and list operations
8. Tuples and immutability
9. Sets and set operations
10. Dictionaries and key-value logic
11. Indexing, slicing, and nested data structures
12. Conditional statements
13. Loops
14. Loop control statements
15. Functions and return values
16. Function arguments
17. Lambda functions
18. Comments and docstrings
19. Basic error handling

INTERMEDIATE LEVEL

20. Reading and writing text files
21. Reading and writing CSV files
22. Reading and writing JSON files
23. File paths and directory handling
24. Context managers
25. Python modules and packages
26. Import styles and best practices
27. Virtual environments and dependency management
28. Working with date and time
29. Object-oriented programming concepts
30. Constructors and instance variables
31. Inheritance and method overriding
32. Encapsulation and class design
33. Pythonic coding practices
34. Comprehensions
35. Built-in utility functions
36. Iterators and generators
37. Writing reusable and modular code

DATA ANALYSIS CORE

38. NumPy arrays and vectorized operations
39. NumPy aggregations and broadcasting
40. Handling missing values with NumPy
41. Pandas Series and DataFrames
42. Reading data into Pandas
43. Indexing and slicing
44. Filtering and sorting data
45. Data type conversions
46. Handling missing and duplicate data
47. GroupBy operations and aggregations
48. Merging and joining DataFrames
49. Window and rolling operations
50. Date-time operations
51. Exploratory data analysis
52. Descriptive statistics
53. Outlier detection and data validation
54. Data visualization with Matplotlib
55. Data visualization with Seaborn

DATA WAREHOUSING CONCEPTS

56. OLTP vs OLAP systems
57. Data warehouse architecture
58. Fact tables and dimension tables
59. Star schema design
60. Snowflake schema design
61. Slowly changing dimensions
62. Surrogate keys and natural keys
63. Grain and level of detail
64. Conformed dimensions
65. Data marts
66. Incremental and full loads
67. Data quality checks
68. Historical data management
69. Partitioning and indexing concepts

ADVANCED PYTHON AND CLOUD

70. Large file processing
71. Memory optimization
72. Working with relational databases
73. SQL integration using Python
74. Parameterized SQL queries
75. ETL concepts
76. Data pipeline design
77. Logging and monitoring
78. Production-level exception handling
79. REST APIs
80. Authentication and pagination
81. API-driven data pipelines
82. Decorators and advanced functions
83. Closures
84. Performance optimization

- 85. Multithreading and multiprocessing
- 86. Asynchronous programming
- 87. Debugging and profiling
- 88. Unit testing frameworks
- 89. Testable data pipelines

BIG DATA AND DISTRIBUTED SYSTEMS

- 90. PySpark architecture
- 91. Spark DataFrames and Spark SQL
- 92. Transformations vs actions
- 93. Window functions in Spark
- 94. Spark performance tuning
- 95. Parquet, Avro, and ORC formats
- 96. Cloud object storage concepts
- 97. Reading and writing cloud data
- 98. Cloud SDKs using Python
- 99. Cloud automation scripts
- 100. Serverless Python basics
- 101. Workflow orchestration fundamentals